

Pratheep Senthilkumar

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SUMMARY

Aspiring software professional with interest in web development, machine learning, and software engineering. Good knowledge of Python and Java with exposure to deep learning concepts and DevOps fundamentals through academic projects and hands-on learning. Strong understanding of core computer science concepts with good problem-solving and analytical skills, along with an eagerness to learn and adapt to new technologies.

EDUCATION

B.Tech Computer Science and Engineering (AI & ML) 2023 – Present
Vellore Institute of Technology, Chennai
CGPA: 9.10

Class XII (CBSE) 2022 – 2023
Akshara Vidyalyaya, Nellore
Percentage: 96.6

SKILLS

- **Programming Languages:** C, C++, Java, Python, JavaScript, SQL
- **Web Technologies:** HTML, CSS, React
- **Core Concepts:** Object-Oriented Programming, DBMS, Operating Systems, Computer Networks, DSA, Machine Learning, Deep Learning
- **Tools & Methodologies:** scikit-learn, DevOps Fundamentals, Agile Methodology

PROJECTS

Brain Tumor MRI Classification using Hybrid Ensemble Learning

- Developed a deep learning-based brain tumor MRI classification system using CNN, InceptionV3, ConvNeXt, and Random Forest models.
- Implemented ensemble learning techniques to improve classification accuracy and model performance on MRI datasets.
- Built a Gradio-based interface for real-time tumor prediction and confidence score visualization using TensorFlow/Keras.

Weather-Wardrobe – DevOps CI/CD Pipeline Implementation

- Developed a weather-based outfit recommendation application using Python and OpenWeather API.
- Implemented CI/CD workflow using GitHub Actions, Docker, and Ansible to automate build and deployment processes.
- Configured monitoring and performance visualization using Graphite and Grafana dashboards.

PUBLICATION

- Co-authored the IEEE SPERT 2025 paper, “**Efficient Deployment of Machine Learning Models for Tidal Current Prediction**”, focusing on tidal current and power forecasting using Random Forest Regressor (RFR) and GRU models.

CERTIFICATES

- **DevOps Fundamentals** — IBM Career Education Program
- **DevOps, Agile & Design Thinking** — IBM Career Education Program